DATA STRUCTURE

LAB PROGRAM

17. write a C program to implement a **Priority Queue using Heap**.

```
#include <stdio.h>

#define MAX 100

int heap[MAX];
int size = 0;

// Function to insert into heap
void insert(int value)
{
    int i = size;

    heap[size] = value;
    size++;

    // Heapify Up (Min Heap)
    while(i != 0 && heap[(i - 1) / 2] > heap[i])
    {
        int temp = heap[i];
        heap[i] = heap[(i - 1) / 2];
```

```
        heap[(i - 1) / 2] = temp;

        i = (i - 1) / 2;
    }
}

// Function to heapify down
void heapify(int i)
{
    int smallest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;

    if(left < size && heap[left] < heap[smallest])
    {
        smallest = left;
    }

    if(right < size && heap[right] < heap[smallest])
    {
        smallest = right;
    }

    if(smallest != i)
    {
```

```
        int temp = heap[i];
        heap[i] = heap[smallest];
        heap[smallest] = temp;

        heapify(smallest);
    }
}

// Function to delete minimum element
void delete()
{
    if(size == 0)
    {
        printf("Heap is empty\n");
        return;
    }

    printf("Deleted element: %d\n", heap[0]);

    heap[0] = heap[size - 1];
    size--;

    heapify(0);
}
```

```
// Function to display heap
void display()
{
   int i;

   if(size == 0)
   {
      printf("Heap is empty\n");
      return;
   }

   printf("Heap elements:\n");

   for(i = 0; i < size; i++)
   {
      printf("%d ", heap[i]);
   }

   printf("\n");
}

int main()
{
   int n, i, value;
```

```
    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &value);
        insert(value);
    }

    display();

    delete();

    display();

    return 0;
}
```

Output

```
Enter number of elements: 5
Enter elements:
555
55
5
555555
5555
Heap elements:
5 555 55 555555 5555
Deleted element: 5
Heap elements:
55 555 5555 555555


=== Code Execution Successful ===
```